# BISHESH KHANAL

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Civil Engineering graduate with first-author research in flood inundation modelling and precipitation extremes. Experienced in HEC-RAS 2D, HEC-HMS, GIS, and Python-based machine learning for hydrological prediction.

## Education

*2022–2026* **TRIBHUVAN UNIVERSITY (TU)** **KATHMANDU, NP**
Bachelor of Civil Engineering, IOE Thapathali Campus, TU

*2020–2022* **PRASADI ACADEMY** **LALITPUR, NP**
High school education

## Research Interests

Flood risk modelling and hydraulic simulation, machine learning surrogates for real-time flood early warning, precipitation and streamflow extremes, and climate-risk assessment in data-sparse mountainous regions.

## Projects & Publications

*2025–2026* **FLOOD INUNDATION MAPPING AND EVACUATION ROUTING: AN APPROACH FOR FLOOD RISK MANAGEMENT** **FINAL YEAR PROJECT**
*(under review, Environmental Development, Elsevier; first author)*
- Developed a hydrologic and hydraulic model and built 2D flood simulation models to analyze spatial flood extent and flow dynamics
- Analyzed the critical infrastructure within the flood hazard zones
- Proposed evacuation strategies based on flood depth and hazard zones

*Aug 2026* **RAINFALL VARIABILITY AND EXTREME EVENT CLASSIFICATION USING MACHINE LEARNING IN THE NARAYANI BASIN, NEPAL**
*(under review, Journal of Water and Climate Change, first author)*

*June 2026* **COMPARING PROCESS-BASED AND MACHINE LEARNING MODELS FOR STREAMFLOW PREDICTION IN THE KALIGANDAKI RIVER BASIN, NEPAL** **EARTHARXIV**
*DOI of preprint:* 10.31223/X55N3N
- Developed and evaluated machine-learning streamflow prediction models in Python
- Applied feature engineering, hyperparameter tuning, and SHAP-based explainability techniques
- Compared process-based and data-driven approaches for hydrological prediction and forecasting

*2025* **FLOOD HAZARD MAPPING AND STRUCTURAL DESIGN RECOMMENDATION** **8TH SEMESTER PROJECT**
- Conducted floodplain surveys along the Hanumante and Manohara river systems to support flood hazard mapping and hydraulic analysis
- Applied the surveyed data to generate terrain for detailed analysis
- Generated inundation maps and applied HEC-RAS and HEC-HMS for hydraulic and hydrologic simulation
- Recommended both engineering and non-engineering measures for reducing flood vulnerability

## Technical Skills

**Programming:** *Python, LaTeX*

**Libraries & Frameworks:** *NumPy, Pandas, Scikit-learn, PyTorch, XGBoost, SHAP*

**Modelling & GIS Tools:** *HEC-RAS 2D, HEC-HMS, SWAT, ArcGIS Pro, QGIS, ArcMap*

**ML & DL Models:** *Random Forest, XGBoost, SVM, LSTM, GRU*

**Domains:** *Flood & Hydrological Modelling, Precipitation Extremes, Water Resources, Remote Sensing & GIS*

**Languages:** *Nepali (native), English, Hindi*

## Certifications & Professional Courses

- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI & Stanford Credential
- **Advanced Learning Algorithms** — DeepLearning.AI & Stanford
- **Neural Networks and Deep Learning** — DeepLearning.AI Credential

## Conferences & Workshops

*Dec 2026* **CO-AUTHOR ABSTRACT, H091 (INTEGRATIVE APPROACHES TO URBAN FLOODING)** **AGU26**

"A Deep Learning Surrogate of a HEC-RAS 2D Model for Urban Flood Forecasting in the Kathmandu Valley, Nepal"

Trained a GRU surrogate of a calibrated HEC-RAS 2D model on 244 synthetic flood events; achieved an overall $R^2$ of 0.947 and mean NSE above 0.88 at a 3-hour lead time. *(abstract accepted, AGU Fall Meeting 2026, San Francisco)*

*Aug 2026* **HANDS-ON TRAINING WORKSHOP ON MEASURING URBAN WATER SECURITY** **LALITPUR, NP**

*ABCD Centre, Asian Institute of Technology (AIT) — with IOE, TU, and CREEW, Nepal*

*June 2026* **PRESENTER, 4TH THAPATHALI GRADUATE CONFERENCE** **THAPATHALI, NP**

"Flood Inundation Mapping and Evacuation Routing: An Approach for Flood Risk Management"

## Honors and Awards

*2026* **BEST FINAL YEAR PROJECT** **IOE THAPATHALI**

Recognised as one of the best final year projects by the department project committee.

*2022–2026* **SEMESTER EXCELLENCE UNDERGRAD AWARD** *(received 6 times)* **IOE, TU**

*References available on request.*